

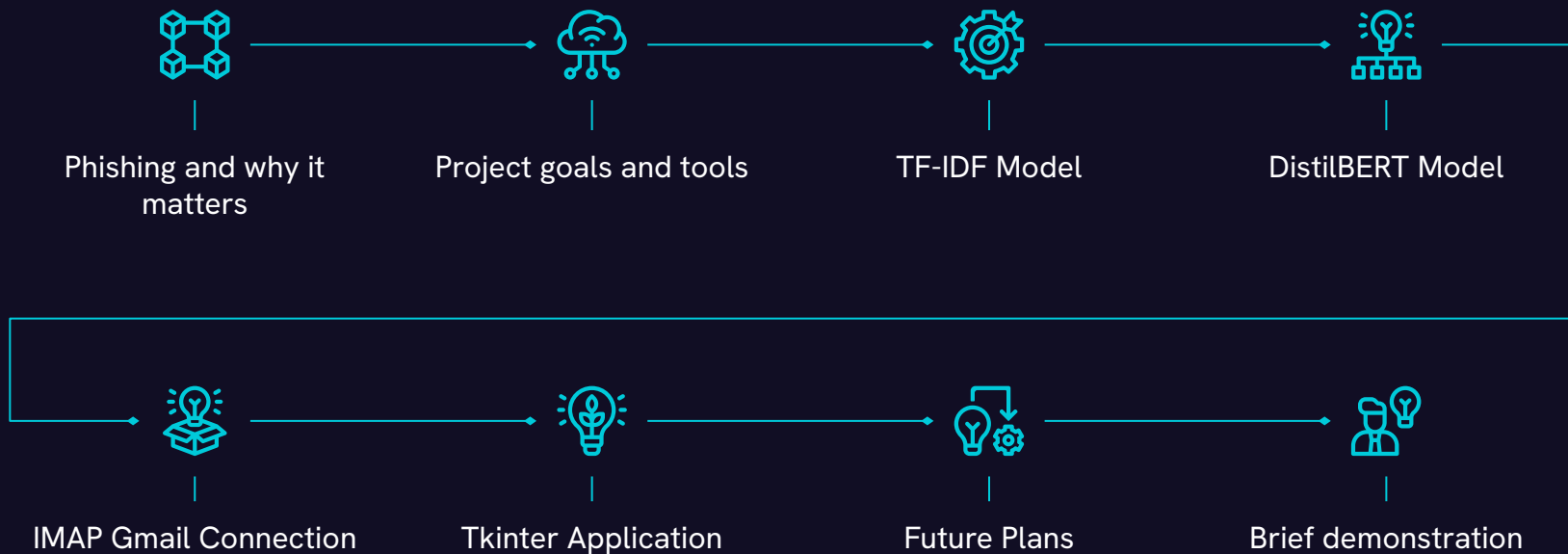


PhishyScanner

Phishing Scanner Application

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► **TIMELINE**



WHAT IS A PHISHING EMAIL?

A phishing email is a bad actor's attempt to gain access to critical information about an individual or a business via malicious links and attachments.

WHY DOES IT MATTER?

- Phishing is evolving, with more realistic emails now.
- It's crucial to protect yourself, attackers mimic trusted services!
- High risk: financial loss, data breaches



► What did we do about it?

Goal: Build an end-end application to detect phishing vs legitimate emails, integrated with real email.



PhishyScanner

We created an application that can connect to your email account, scan new emails, and accurately assess them as legitimate or malicious.



OUR IDEA

The use of our trained model allows us to score each individual email through various factors such as:

- Use of urgent language.
- Untrusted links.
- Malicious attachments

Placing them into three risk tiers:

- Safe
- Suspicious
- Likely Phishing

► What tools are being used?



Pycharm

Python IDE where our model is being trained.

Tech Stack:

- Scikit-learn (TF-IDF)
- Hugging Face Transformers (DistilBERT) - in testing



IMAP

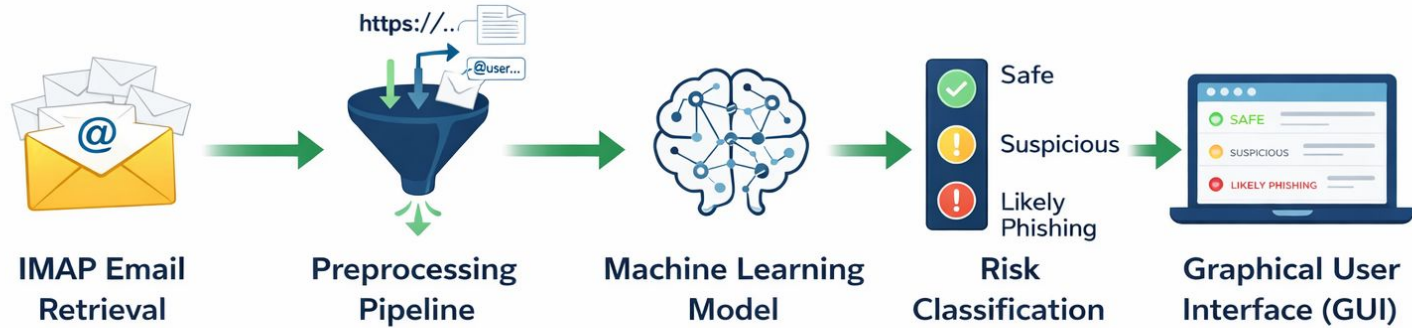
IMAP is a connection medium for us to get direct access to a user's email, allowing the model to scan for threats.



TKINTER

Tkinter is the python native platform we used to create an application window for a user interface for our model.

Data Sources: Kaggle phishing datasets (~80k emails), custom modern dataset (800 emails), Phishing validation email dataset from Zenado (2,000 emails).
Modern datasets are crucial for the model to understand realistic email scenarios!



► **MODEL Version 1: TF-IDF -> Strong baseline**

What it does: converts text into numerical features. Uses word and character n-grams, to help the model capture some local context and structural info. Trained with logistic regression → Fast and interpretable baseline.

Test Set Results (rounded):

Accuracy: 98%

Precision: 99%

Recall: 96%

Strong performance on the dataset. Sample detections:

- Replica watch spam
- Pharmacy emails
- Urgent verification emails

Limitations:

- Heavily influenced by older spam data, and had limited improvement after modern dataset
- Relies heavily on keywords, not full context
- Struggled with some realistic phishing emails

► **MODEL Version 2: DistilBERT**

What is it? A transformer based model, that understand language context. Pre trained on large text corpora and fine tuned for phishing classification.

Why the switch? More adaptable to modern threats. Captures full meaning!! Great for security alerts, document sharing emails, realistic phishing attempts.

First training: Achieved 99% test accuracy and reduced the number of missed phishing emails from 284 with TF-IDF to 52 with DistilBERT.

Threshold Tuning: Initial approach used precision/recall based thresholds, which produced unrealistic values. Switched to a percentile based method using calibration data. Defined thresholds using both legitimate and phishing score distribution to make more stable tiers:

- Safe
- Suspicious
- Likely Phishing

Results: More consistent risk scoring!

Currently: Doing more tuning before deployment to the app.

► IMAP GMAIL CONNECTION



How it works

Our imap code connects to a given gmail account through their provided email and *Gmail App Password*



Example password

kwey jfoi nvhv hsal



preprocess

Grabs unread emails from the inbox, taking metadata, subject, and message body, then enters it into the preprocess.py



Analyze

The emails are fed into our model and scored for phishing similarities.



Addresses issue

The imap moves the detected phishing emails from the main inbox, into the users spam inbox.

► TKINTER APPLICATION

User interface



Login

Requests the users email and their *Gmail App password*



Dashboard

Shows all scanned emails:

- How trusted they seem
- Their trustworthiness score .00 - 1.00



Analytics

A summary report of the distribution of each tier of email received.

A number report of emails quarantined.



What still needs to be done?

Our plans from now until the final presentation

▶ Gmail Overlay

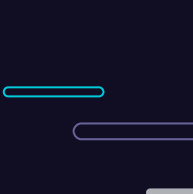
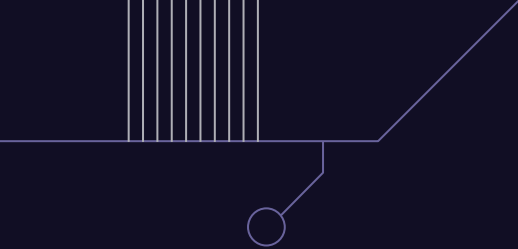
Browser extension + Improve UI

▶ Batch system for retraining

Retraining based on real emails from user

▶ Security Measures

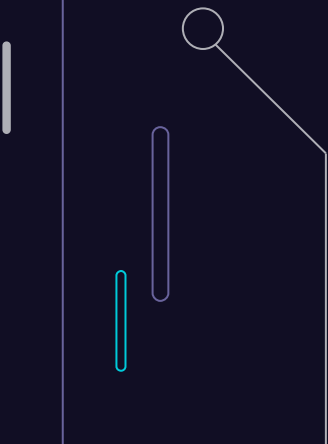
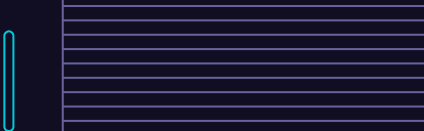
Reduce risk of data breach and implement privacy steps



Demo

TF IDF MODEL





► **THANK YOU**

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